OutOfRole: Build Plan

The Debugging Table · IBM Z Datathon 2026 · October 17 to 18

What we're building

OutOfRole spots insiders while they are still active, explains each alert with checked evidence, and suggests which access each person should keep pre-approved. It all runs on one LinuxONE server, and people appear only as pseudonyms (U-017).

1. **Early warning (the headline).** Replay the logs day by day. Compare each person with their own normal weeks; the 20 most unusual people form a daily review list.
2. **Granite case files.** For each person on the list, code ranks the evidence ("new to this person, rarest first"). Granite picks a label from a fixed list and the supporting facts; code writes every fact sentence and rejects any other reply.
3. **Least-privilege advisor (the recommender).** A recommender predicts which websites each person is likely to use next. Those can be pre-approved; anything else goes to a reviewer. Sites no one in the company visited during the training months are blocked.

Pitch line: "In a day-by-day replay of 2,000 employees' logs, OutOfRole put 80 of 99 insiders on a 20-name daily review list while their attacks were under way, twice as many as a standard 28-day baseline on the same 20-name list, and on average about half of each one's malicious actions came after the flag. Granite, on the same server, labels each case and picks the supporting evidence, which code writes into a checked case file. A recommender suggests which access to pre-approve, so reviewers see fewer routine requests."

Level by level: what happens and what it buys us

Each level runs on the output of the one above it. Grey bars show the reference (chance or the previous best), and teal bars show the level. Release r4.2 has 70 insiders among 1,000 staff; r5.2 has 99 among 2,000. “Chance” at L1 is the expected number of insiders in 100 randomly chosen people.

DATA

L0
Logs to tables

Logon, USB, file and web logs (15 GB for r4.2, 37 GB for r5.2) become three Spark tables per person per day.

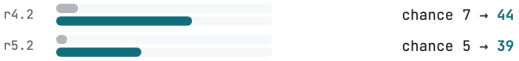
r4.2 alone has 1,003 computers and 475 websites. The web log moved from a stretch goal to core, because rare-site visits turned out to be a strong signal.

DETECT

L1
Personal baseline

Is this person's last 28 days unusual compared with their own history?

Insiders found when an analyst reviews the top 100 people



L2
Own-history weekly percentile

Did this week beat every one of the person's normal weeks, on any activity?

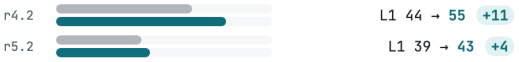
Real insiders among the top 20 names



L3
Signal families combined

Behaviour, own history, peers, rare sites and slow drift, fused without labels (family-first Stouffer).

Insiders found in the top 100 people



End-of-period snooping review, built on the recommender: weeks with logins to computers outside the person's profile put 27 of 30 r5.2 snoopers in its top 100 (14 in the top 20); without the recommender, 26. As a live list it catches 9 of 30 in time, so live warnings come from the daily list.

WATCH DAILY

L4
Early warning

Replay the logs one day at a time, using only the past. The 20 highest running scores form that day's review list.

Insiders on the list before their last malicious action



Grey is the 28-day baseline on the same 20-name list. Under our running rule it gets only 33 and 28, and a random score gets 10 and 11. On average, about half of each insider's labelled malicious actions (54%, 48%) came on later days. Median delay: 0 days for leakers (most were already listed) and 1.5 to 3 weeks for job-hunters and snoopers. Analyst load is about 5 new names a week.

EXPLAIN

L5
Evidence and Granite case file

For each flagged person, rank their activity "new to this person, rarest first". Granite picks a label and the supporting facts; code writes every fact sentence.

Insiders flagged in time with a real malicious event in their top 3 items (evidence up to the flag day)



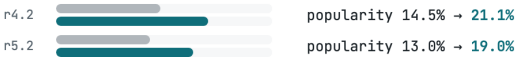
Leakers, sysadmins and snoopers: 85 to 100%. Job-hunters are the hard case. The labels are used only to grade, never to rank.

PREVENT

L6
Least-privilege advisor

The recommender predicts each person's next websites, so those can be pre-approved; anything else goes to a reviewer. Sites no one here visited during the training months are blocked.

Next new websites caught by the top 20 guesses (bar scale 0 to 30%)



Access requests drop from 2.31 to 1.82 per person-month (3.15 to 2.55 on r5.2). It suggests rather than decides: only 28% and 42% of its extra grants were later used. Removal suggestions: 6.0% and 3.6% of the sites people hold go unused afterwards. The new-to-company rule blocks 100% of wikileaks and keylogger visits, costing 5 normal people 8 days in total on r4.2.

Numbers we tested

CERT data. Settings were frozen on r4.2, then r5.2 was run unchanged. Each number was reproduced with independent code.

Claim	r4.2 (70 insiders)	r5.2 (99 insiders)
Insiders on the daily list before their last malicious action	**59**	**80**
Same, 28-day personal baseline on the same 20-name list (the fair comparison)	59	40
Same, 28-day baseline under our running rule (the pre-registered comparison)	33	28
Same, by chance (shuffled real list histories)	about 3	about 3
Same, a random score on the same 20-name list	10	11
Average share of each insider's labelled malicious actions on days after the flag	54%	48%
New names per week for the analyst	about 5	about 5
Insiders flagged in time with a real malicious event in their top 3 evidence items, using evidence up to the flag day (global rarity in brackets)	64% (49%)	85% (51%)
Next new websites caught by the recommender's top 20 (popularity in brackets)	21% (14.5%)	19% (13%)
Access requests per person per month: history only, then with the recommender	2.3 to 1.8	3.15 to 2.55
Held sites never used again (removal suggestions for access admins)	6.0%	3.6%
Wikileaks and keylogger visits blocked by the new-to-company rule	100%	100%
Snoopers in the top 100 of the end-of-period snooping review (without the recommender in brackets)	no snoopers	27 of 30 (26)
Same review as a live 10-name daily list: snoopers caught in time (daily list alone: 25)	no snoopers	9 of 30
Strong evidence at first alert, insiders vs normal staff (no labels used)	29% vs 18%	65% vs 31%

By attack type:

- Leakers: median delay 0 days from the first labelled malicious day, mostly because they were already on the list. Only 7 of 30 and 15 of 29 were newly flagged.
- Job-hunters and snoopers are flagged after 1.5 to 3 weeks.
- Two-day sysadmin sabotage is mostly missed: 4 of 10 on r4.2, 2 of 10 on r5.2.

Step by step: what each test changed

"Found" = insiders expected in the top 70. Full detail: the notebook and `5_Evidence/Verified_Numbers.md` section 7.

Step	What we tried	Metric	Before	After	Change	Verdict
3	Personal 28-day baseline	found, r4.2 / r5.2	none	38 / 36	the bar	bar
4	Rare-site visits	found, r4.2	38	41	+3 (r5.2: 8.1)	one family
5, 6	Trained combiners	found, r4.2	39	45 to 60	+6 to +21	fail on new attacks
7	Label-free sum, job-hunters hidden	job-hunters in top 100	4 to 7	16 of 30	+9 to +12	label-free wins
8	Weekly LOF (literature)	job-hunters in top 100	5	12 to 13	+7 to +8	slow drift
10	Own-history weekly percentile	found, r4.2	38	39.4, P@20 1.00	+1.4	core
11	Family-first Stouffer fusion	found, r4.2	38	45	+7	frozen
12	Same pipeline on r5.2, frozen	found, r5.2	36	38	+2	survives
13	Snooper persistence signal	snoopers in top 100	10.9	27 of 30	+16	view
14	Exactly 100 reviews	insiders, r4.2 / r5.2	44 / 39	55 / 43	+11 / +4	default ranking
16	Recommender, tuned without labels	next-site recall@20, r4.2	0.145 (popularity)	0.211	1.45x	real job
17	Least-privilege advisor	requests per person-month	2.31 / 3.15	1.82 / 2.55	-21% / -19%	build
18	Evidence rule for Granite	hit@3, insiders flagged in time	0.49 / 0.51 (rarity)	0.64 / 0.85	+0.15 / +0.34	build
19	Early warning, day by day	flagged in time, r4.2 / r5.2	59 / 40 (baseline, same 20-name list)	59 / 80	0 / +40	headline
20	Analyst feedback; richer recommenders	insiders in 100 reviews	55 / 43	57 / 42	noise	dropped

Steps 1, 2, 9 and 15 were checks or were superseded:

- Step 1: data loading.
- Step 2: the leakage check.
- Step 9: recommender tweaks (peer-of-item: 17 found).
- Step 15: the untuned recommender verdict, replaced by Step 16.

Who builds what

Fill in names. Hours count from the start of our build.

Role	Hours 0 to 6	Hours 6 to 12	Hours 12 to 18	Hours 18 to 24
R1 Data and platform (_____)	VM: Java 17, PySpark (no pyarrow on s390x), Docker; download r4.2. At hour 2, hand a one-month sample to R2 and R4. Spark builds 3 tables: daily counts, person-computer-day, person-site-day	r5.2 if disk allows; run the whole pipeline end to end	Docker container, GitHub repo, README	Support the demo; check the repo runs from scratch
R2 Detection (_____)	Port section 1 of `outofrole_minimal.py` (the daily list) and run it on the sample	Run on r4.2 with the fixed settings below	r5.2 unchanged; results table vs baseline and chance, per attack type, analyst load	Export daily lists and per-person timelines for the dashboard
R3 Granite (_____)	Both IBM labs (Granite via llama.cpp; the model file must be big-endian)	Port `case_items` from `outofrole_minimal.py` and all of `case_file.py` (evidence pack with IDs, Granite prompt that returns only a label and fact IDs, reply check, fallback); pseudonyms	Pre-generate case files for the top 20; keep one live run	Case-file panel in the dashboard
R4 Recommender and demo (_____)	Dashboard skeleton: daily list, person page, case-file panel	Port sections 3 to 6 of `outofrole_minimal.py`: ALS on person-site data, top-20 pre-approval, removal suggestions, new-to-company block, and the recommender snooping review (end of period)	Wire everything into the dashboard	Video with on-screen proof: `uname -m` (s390x), `docker ps`, the server prompt

Fixed detector settings (decide nothing new):

- A feature is unusual when the last 7 days beat every one of that person's training weeks.
- The running score halves every 7 days.
- The list holds 20 names a day.

Recommender settings: rank 64, alpha 0.5, regularization 1.0 on log(1 + active days), using the numpy version in `recommender_als_simple.py`. Spark's ALS scales regularization differently, so if you use it, re-tune on the last training month and pick by how many next-month new sites it catches.

Cut rules

- **Hour 6:** if the Spark tables aren't ready, run r4.2 with logon, USB and file data only; add web later.

- **Hour 12:** freeze the detector settings. No tuning after this.
- **Hour 14:** if the recommender is behind, ship the new-to-company block plus each person's own history as the allow-list. Cite the recommender's local result instead.
- **Hour 18:** if Granite is slow or unreliable, show the pre-generated case files plus one live run. If it fails entirely, show the checked evidence pack itself.
- **Hour 20:** feature freeze. Only the video, README and slides from here.

Demo (3 minutes)

1. The replay plays. A leaker joins the daily list on the day of the first wikileaks upload.
2. Click the person. The case file appears: Granite's label, the top-ranked evidence written by code, and the facts Granite cited marked. Then show a bad reply being rejected and replaced by the fallback.
3. Access panel: 20 pre-approved sites. A request for wikileaks.org is blocked as new to the company.

Say this, not that

- Say "flags insiders while they are still active", not "real-time production system". It is a replay of past logs.
- Say "the recommender predicts legitimate access", not "the recommender detects insiders". We tested that: as a detector it is no better than popularity.
- Say openly that two-day sysadmin sabotage is mostly missed, and that CERT is synthetic data.
- Say "suggests which access to pre-approve", not "trims access to what people need". Only 28% (r4.2) and 42% (r5.2) of the recommender's extra grants were later used.
- Say plainly that the approach changed after testing. The abstract promised recommender-based detection; our tests showed that own-history baselines detect better, so the recommender moved to access advice.
- Compare against the strongest baseline. On the same 20-name list, the 28-day baseline flags 59 of 70 (a tie on r4.2) and 40 of 99 (half of ours on r5.2).
- The early-warning method was picked after seeing results. The full detector we had named in advance also beat the baseline: 54 vs 28 of 99 r5.2 insiders flagged in time.

Reference code (we rewrite it on the server)

In `4_Code/`, in the order they run:

File	What it does	Who ports it
`build_tables.py`	Spark turns the raw logs into the three tables (daily counts, person-computer-day, person-site-day)	R1
`outofrole_minimal.py`	The whole system, printing every headline number: daily list (section 1, lines 43 to 94), snooping review (1b, lines 97 to 122), evidence ranking (2, lines 124 to 186), recommender and access advice (3 and 4, lines 188 to 215), new-site rule (5, lines 217 to 221)	R2: sections 1 and 2; R4: 1b, 3 to 5
`case_file.py`	Evidence pack, Granite prompt, reply check and fallback; run it to see the self-test	R3
`checks/` `followup_checks.py`	Resampling, rolling baseline, role changes, monthly retraining	optional
`experiments/`	The original study scripts behind the evidence	reference only

- Full study with code: `5_Evidence/Study_Notebook.ipynb`
- Data download steps: `Data_Download_Guide.pdf` (this folder)